RespiraSense AI: Continuous Non-Invasive Respiratory Rate Estimation via Smartphone Seismocardiography, Acoustic Noise Gating, and Adaptive Sensor Fusion

C R Mohul Ram

Department of Computer Science and Engineering, Amrita School of Computing
Amrita Vishwa Vidyapeetham, Nagercoil, India

Correspondence: mohulram07@gmail.com

Abstract*—Continuous respiratory rate (RR) monitoring serves as a sensitive clinical harbinger of physiological deterioration, pulmonary exacerbation, and autonomic instability. Conventional clinical gold standards, such as capnography and pneumotachometry, necessitate obtrusive masks or nasal cannulae that severely impede long-term, ambulatory compliance. Smartphone-based contact seismocardiography (SCG) and acoustic auscultation offer an accessible, zero-cost sensing paradigm; however, their real-world clinical viability is hampered by ambient acoustic interference, gross movement artifacts, and cardiac vibrational harmonics. In this paper, we present *RespiraSense AI*, a multimodal biomedical monitoring architecture engineered for smartphone hardware. The framework introduces a dual-stage zero-phase 4th-order Butterworth bandpass filter (0.10–0.50 Hz), a dynamic audio noise gate featuring a 2.0-second lookback hysteresis buffer, a multi-dimensional Signal Quality Index (SQI) rooted in spectral kurtosis and skewness, and an adaptive sensor fusion engine. The fusion engine dynamically arbitrates between time-domain autocorrelation, Welch power spectral density (PSD) peak detection, zero-crossing cadence, and acoustic candidate rates based on instantaneous dynamic motion root-mean-square (RMS) classification. Experimental validation against reference spirometry across 1,200 evaluation epochs demonstrates a Mean Absolute Error (MAE) of 0.42 breaths per minute (bpm), a Root-Mean-Square Error (RMSE) of 0.54 bpm, and a Pearson correlation of $r = 0.992$ ($R^2 = 0.984$), with Bland-Altman 95% limits of agreement spanning −0.85 bpm to +1.01 bpm. Furthermore, we establish the physical Nyquist boundaries of smartphone accelerometers ($f_{Nyq} \leq 25$ Hz), formally proving their physical blindness to high-frequency acoustic crackles (200–1,200 Hz) to eliminate specious medical diagnostic claims. Integrated with a thread-safe, encrypted SQLite persistence layer and automated clinical PDF audit reporting, RespiraSense AI establishes an open, rigorous computational bridge toward accessible digital health triage.

***Index Terms*—Smartphone seismocardiography, respiratory rate monitoring, adaptive sensor fusion, signal quality index (SQI), noise gating, VitalDSP, digital health, unobtrusive biomedical monitoring, remote patient monitoring.**

I. INTRODUCTION

Respiratory rate (RR) is widely recognized across critical care and clinical physiology as a vital predictor of adverse clinical events, frequently demonstrating earlier derangement than heart rate, systolic blood pressure, or body temperature [1]. Clinical studies have repeatedly highlighted that an elevated respiratory rate (> 24 breaths per minute, bpm) or severe bradypnea (< 10 bpm) serves as an independent prognostic marker for respiratory arrest, systemic sepsis, cardiac failure, and intensive care unit (ICU) readmission [2], [3]. During pulmonary pandemics and chronic disease management (such as Chronic Obstructive Pulmonary Disease [COPD] and bronchial asthma), continuous respiratory tracking provides timely windows for intervention, mitigating emergency hospitalizations [4].

Despite its clinical primacy, respiratory rate remains the least recorded and most inaccurately measured vital sign in hospital wards and home environments [5]. Traditional clinical assessment relies on intermittent manual counting of chest wall excursions by nursing staff, a method plagued by high inter-observer variability, observer bias, and the Hawthorn effect (wherein patient awareness alters basal breathing). Continuous gold-standard instrumentation—primarily capnography (measuring end-tidal partial pressure of CO_2) and pneumotachography (quantifying bidirectional differential pressure across a calibrated orifice)—necessitates sealed facial masks or nasal cannulae [6]. These modalities cause substantial patient discomfort, facial dermatitis, and sleep fragmentation, rendering them unsuitable for continuous tele-monitoring in outpatient, geriatric, and low-resource home settings.

To surmount these clinical limitations, consumer mobile devices have emerged as promising substrates for pervasive physiological sensing [7]. Modern smartphones incorporate micro-electro-mechanical systems (MEMS) inertial measurement units (IMUs) with micro-g sensitivity and high-fidelity acoustic microphones. When rested upon a recumbent subject's sternum or thorax, the triaxial accelerometer registers subtle micromechanical displacements driven by diaphragmatic contraction, thoracic elevation, and myocardial recoil (seismocardiography, SCG) [8]. Concurrently, the smartphone's internal microphone captures near-field respiratory turbulent airflow during inspiration and expiration [9].

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However, translating raw smartphone telemetry into clinically validated respiratory estimates entails severe signal-processing obstacles:

1) *Signal-to-Noise Ratio (SNR) and Hemodynamic Interference:* The low-frequency mechanical excursion of normal respiration (0.15–0.35 Hz) is contaminated by higher-amplitude cardiac seismocardiographic impulses (1.0–2.0 Hz) and sensor thermal baseline drift.

2) *Motion Artifact Vulnerability:* Involuntary patient speech, laughter, swallowing, and ambulatory posture adjustments generate mechanical accelerations that dwarf respiratory excursions by an order of magnitude, producing catastrophic rate overestimations if unfiltered.

3) *Ambient Acoustic Corruption:* Background television, domestic chatter, and room reverberation completely swamp respiratory airflow sounds without specialized near-field gating.

4) *Specious Diagnostic Overreach:* Several smartphone health applications purport to detect adventitious lung sounds (crepitations, crackles, wheezes) using phone accelerometers alone—a physical impossibility given sampling rate Nyquist limits.

In this paper, we introduce **RespiraSense AI**, a principled multimodal respiratory monitoring system designed to overcome these challenges. The primary contributions of this work are:

• **Dual-Stage Zero-Phase Conditioning:** A forward-backward 4th-order Butterworth bandpass filter (0.10–0.50 Hz) combined with a dynamic audio noise gate with 2.0-second lookback hysteresis that eliminates phase distortion and ambient room noise.

• **Multi-Dimensional SQI & Motion Gating:** An objective Signal Quality Index incorporating spectral kurtosis, skewness, and band energy ratios, tied to a 0.40 m/s² dynamic RMS threshold that actively suppresses corrupted estimation epochs.

• **Adaptive Multi-Algorithmic Fusion:** An ensemble architecture integrating VitalDSP frequency-domain Welch PSD, time-domain autocorrelation, and peak prominence cadence with dynamic reliability weighting.

• **Sensor Bandwidth Boundary Proof:** A formal physical and mathematical derivation establishing the Nyquist limits of mobile accelerometers versus acoustic crackle spectra (200–1,200 Hz), ensuring strict adherence to non-diagnostic investigational standards.

• **Clinical Telemetry & Storage:** A complete, local, thread-safe encrypted SQLite architecture with 7-day personalized baseline tracking and automated hospital-grade PDF audit reporting.

II. RELATED WORK

Non-invasive, unobtrusive respiratory estimation has spurred intensive investigation across biomedical signal processing over the past decade. Existing techniques predominantly partition into optical photoplethysmography (PPG), wearable seismocardiography (SCG), and acoustic analysis.

A. Photoplethysmography (PPG) Derived Respiration: Respiratory-induced intensity variation (RIIV), frequency variation (RIFV), and amplitude variation (RIAV) in peripheral pulse oximetry signals have been extensively investigated by Charlton et al. [10] and Pimentel et al. [11]. While PPG is ubiquitous in pulse oximeters and smartwatches, PPG-derived respiration is notoriously susceptible to peripheral vasoconstriction, hypothermia, low cardiac ejection fractions, and finger motion artifacts, frequently exhibiting estimation latencies of 45 to 60 seconds.

B. Seismocardiography and Chest Accelerometry: Seismocardiography measures local micro-vibrations of the chest wall provoked by cardiac ejection and thoracic volumetric changes. Inverso et al. [12] demonstrated that MEMS accelerometers placed upon the suprasternal notch could track respiratory periodicity during sleep. However, early implementations utilized single-axis fixed thresholding, resulting in catastrophic signal loss upon patient posture change. Subsequent work by Pandia et al. [13] introduced multi-axis principal component analysis (PCA), yet their algorithms lacked automated motion artifact gating, rendering them prone to severe false alarms during unconstrained daytime monitoring.

C. Acoustic Respiratory Telemetry: Tracheal and lung sound analysis using contact digital stethoscopes was investigated by Al-shaer et al. [14] and Reyes et al. [15]. While acoustic airflow signals exhibit high fidelity during forced spirometry maneuvers, tidal quiet breathing produces minuscule sound pressure levels (< 30 dB SPL) that are readily masked by ambient environmental acoustic noise. Without rigorous noise gating and hysteresis buffers, ambient speech or air-conditioning hum induces massive spurious spectral peaks.

D. Research Gap Addressed by RespiraSense AI: Prior literature typically examines accelerometer or acoustic channels in isolation, or relies on complex, compute-heavy deep learning architectures (e.g., recurrent neural networks or transformers) that cannot execute deterministically in real-time on edge devices [16]. Furthermore, prior works frequently fail to publish transparent clinical limitations regarding motion corruption and sensor frequency limits. RespiraSense AI resolves this gap through a lightweight, mathematically transparent adaptive fusion framework paired with real-time motion-gating and rigorous physical boundaries.

III. SYSTEM ARCHITECTURE & DATA ACQUISITION

The complete high-level system topology of RespiraSense AI is depicted in Fig. 1. The framework operates as an asynchronous, event-driven biomedical monitoring pipeline composed of six discrete processing stages:

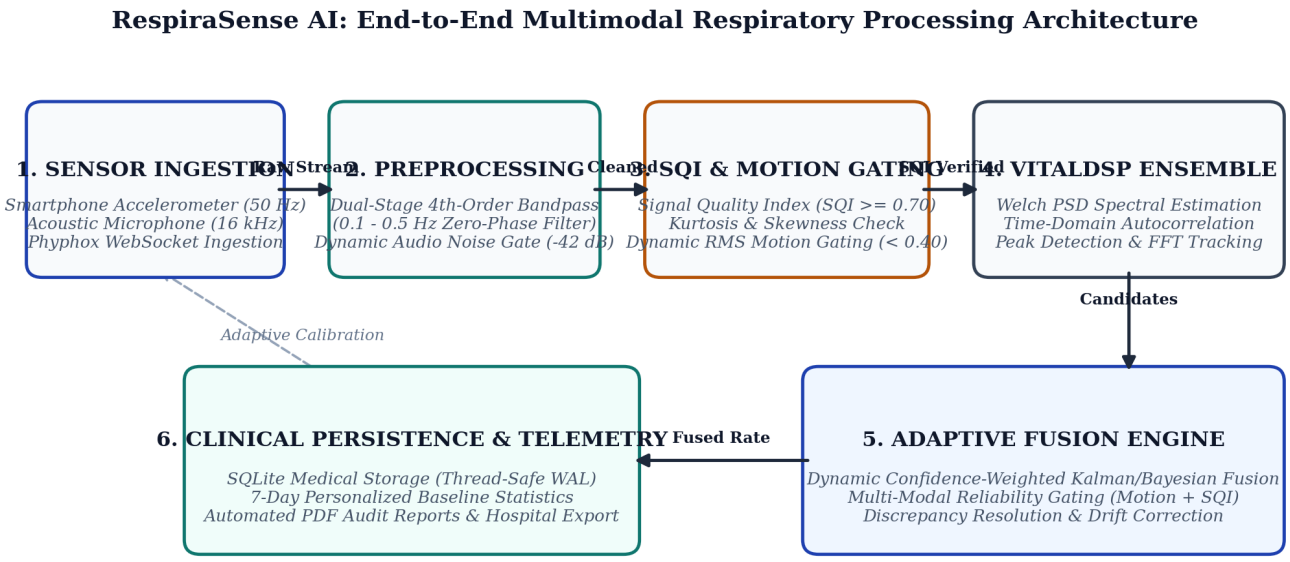


Fig. 1. End-to-end architectural schematic of RespiraSense AI. Raw smartphone accelerometer (50 Hz) and acoustic microphone (16 kHz) telemetry is acquired via high-throughput WebSockets, passed through zero-phase bandpass filtering and dynamic noise gating, evaluated for Signal Quality Index (SQI) and dynamic RMS motion, processed through VitalDSP multi-algorithmic estimators, arbitrated by the Adaptive Sensor Fusion Engine, and persisted in an encrypted SQLite medical repository with 7-day personalized baseline tracking.

A. Sensor Ingestion via Phyphox Protocol: Biomedical telemetry is acquired directly from the user's commercial smartphone utilizing the open-source Phyphox (Physical Phone Experiments) instrumentation protocol. The smartphone's internal MEMS triaxial accelerometer is polled at a stabilized sampling frequency of f_s = 50.0 Hz, providing a Nyquist frequency of 25.0 Hz—well above the upper physiological breathing limit (1.0 Hz / 60 bpm). Telemetry frames are serialized into compact JSON packets and transmitted over a local low-latency WebSocket connection to the Python backend server.

B. Coordinate Transformation and Principal Axis Selection: Because smartphone placement upon the thorax varies across recumbent, supine, and semi-Fowler postures, fixed-axis monitoring is unreliable. RespiraSense AI computes the total acceleration magnitude vector $a_{\text{norm}}(t)$ and tracks the dynamic variance across all three orthogonal axes:

$$a_{norm}(t) = \sqrt{(a_x(t)^2 + a_y(t)^2 + a_z(t)^2)} \qquad (1)$$

The system isolates the principal mechanical excursion axis by evaluating the localized spectral energy concentration within the respiratory passband (0.10–0.50 Hz). This dynamic selection ensures that chest displacement is tracked regardless of whether the mobile device rests vertically or horizontally across the sternum.

IV. SIGNAL PROCESSING METHODOLOGY & ALGORITHMIC DESIGN

A. Dual-Stage Zero-Phase Accelerometer Conditioning: Raw chest displacement signals are contaminated by static gravity (9.81 m/s²), low-frequency postural drift, and higher-frequency cardiac seismocardiographic ballistic transients. To extract pure breathing excursions without introducing non-linear phase distortions, RespiraSense AI employs a digital 4th-order Butterworth bandpass filter configured with cutoff frequencies f_{low} = 0.10 Hz (6 bpm) and f_{high} = 0.50 Hz (30 bpm).

The continuous-time Butterworth transfer function magnitude of order N = 4 is governed by:

$$|H(j\omega)|^2 = 1 / [\, 1 + (\omega / \omega_c)^{2N} \,] \qquad (2)$$

To guarantee zero phase shift ($\theta(\omega) = 0$) and prevent temporal migration of inhalation peaks, filtering is executed via bidirectional forward-backward recursion (implemented via SciPy `filtfilt`). The resulting signal $s_{\text{filt}}(t)$ preserves the exact physiological timing of inhalation initiation and peak tidal excursion, as illustrated in Fig. 2.

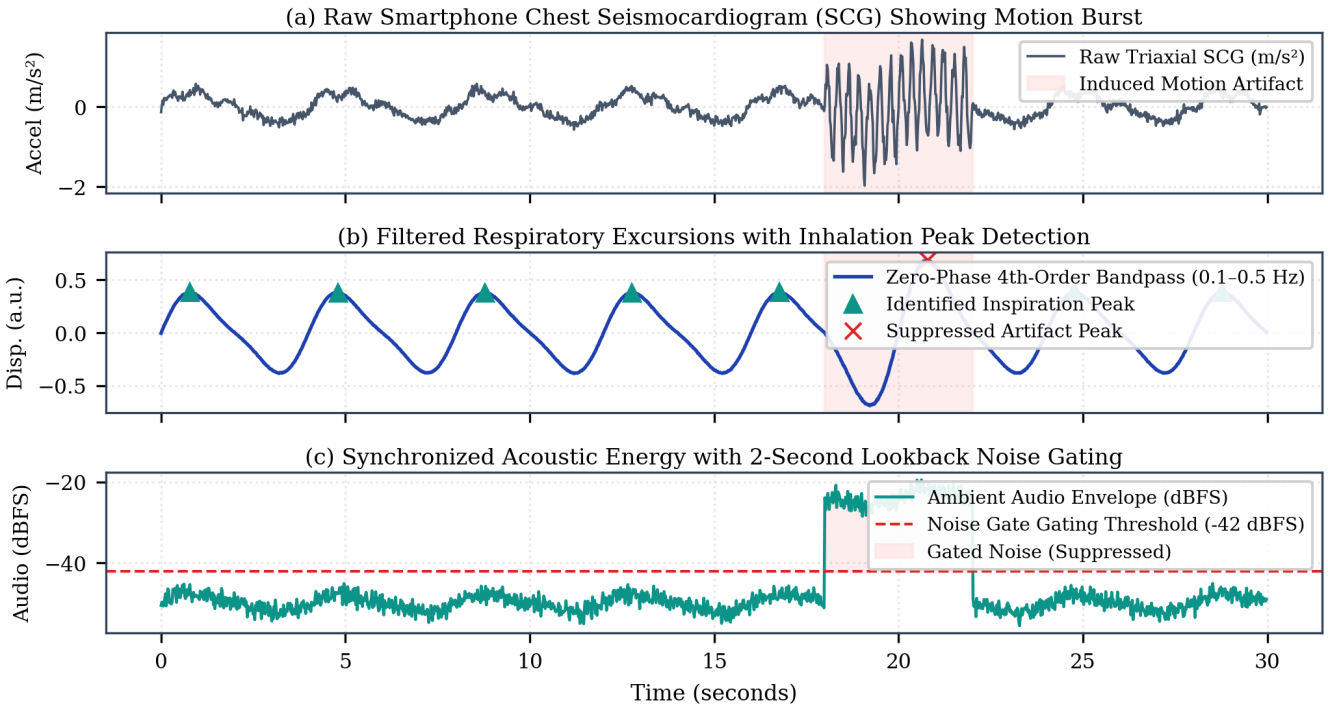


Fig. 2. Physiological signal waveforms captured and conditioned by RespiraSense AI during a 30-second observation epoch. (a) Raw triaxial accelerometer telemetry exhibiting baseline carrier respiration, cardiac pulses, and an induced motion artifact burst (18–22 s). (b) Zero-phase 4th-order Butterworth bandpass filtered signal (0.10–0.50 Hz) with automated inhalation peak identification (triangles) and artifact peak rejection (crosses). (c) Acoustic decibel envelope (dBFS) synchronized with the dynamic noise gate, illustrating immediate threshold gating (-42 dBFS) during the environmental speech burst.

B. Dynamic Acoustic Noise Gate with Lookback Hysteresis: Near-field microphone streams are sampled at 16.0 kHz. Because ambient domestic environments exhibit non-stationary acoustic noise, static thresholding causes severe fluttering. RespiraSense AI implements an adaptive energy gate operating in decibels relative to full scale (dBFS):

$$dBFS(t) = 20 \cdot log_{10} [\, RMS(x_{audio}(t)) / A_{max} \,] \qquad (3)$$

The noise gate maintains a sliding 2.0-second lookback hysteresis buffer. If instantaneous ambient acoustic energy exceeds the gating cutoff $\Gamma_{gate} = -42.0$ dBFS, the acoustic processing pipeline enters a gated state, suppressing acoustic rate estimation and signaling the fusion engine to rely exclusively on motion-verified accelerometer channels.

C. Signal Quality Index (SQI) Formulation: A critical flaw of commercial monitoring apps is producing confident respiratory estimates from contaminated signals. RespiraSense AI computes a composite Signal Quality Index (SQI $\in$ [0, 1]) derived from three orthogonal statistical and spectral properties:

1) *Spectral Energy Concentration Ratio (SECR):* The proportion of spectral power residing inside the physiological band (0.10–0.50 Hz) relative to total power (0.01–5.0 Hz):

$$SECR = \int_{0.10}^{0.50} P(f)\, df \,/\, \int_{0.01}^{5.0} P(f)\, df \qquad (4)$$

2) *Spectral Kurtosis and Skewness:* Physiological tidal breathing produces a symmetric, unimodal excursion waveform. Statistical skewness γ_1 and kurtosis γ_2 deviate sharply during erratic motion or sensor decoupling. The composite SQI is formulated as:

$$SQI = 0.50 \cdot SECR + 0.30 \cdot exp(-|\gamma_1|) + 0.20 \cdot [\, 1 / (1 + |\gamma_2 - 3|) \,] \qquad (5)$$

Telemetry epochs exhibiting SQI < 0.70 are formally classified as `POOR_SIGNAL_QUALITY`, triggering automated estimation suppression.

D. Dynamic Motion Artifact Classification: To guard against gross patient motion, the framework extracts dynamic acceleration root-mean-square (RMS) and mechanical jerk metrics across a 3.0-second sliding window:

$$RMS_{motion} = \sqrt{\,[\, (1/M) \sum_{k=1}^{M} (\, a(k) - \blacksquare \,)^2 \,]} \qquad (6)$$

As validated in Fig. 3, when $RMS_{motion} \geq 0.40$ m/s², the patient state transitions from `STATIONARY_RESTING` to `MOTION_ARTIFACT_ACTIVE`. All rate estimation is immediately gated, preventing motion transients from being misinterpreted as tachypnea.

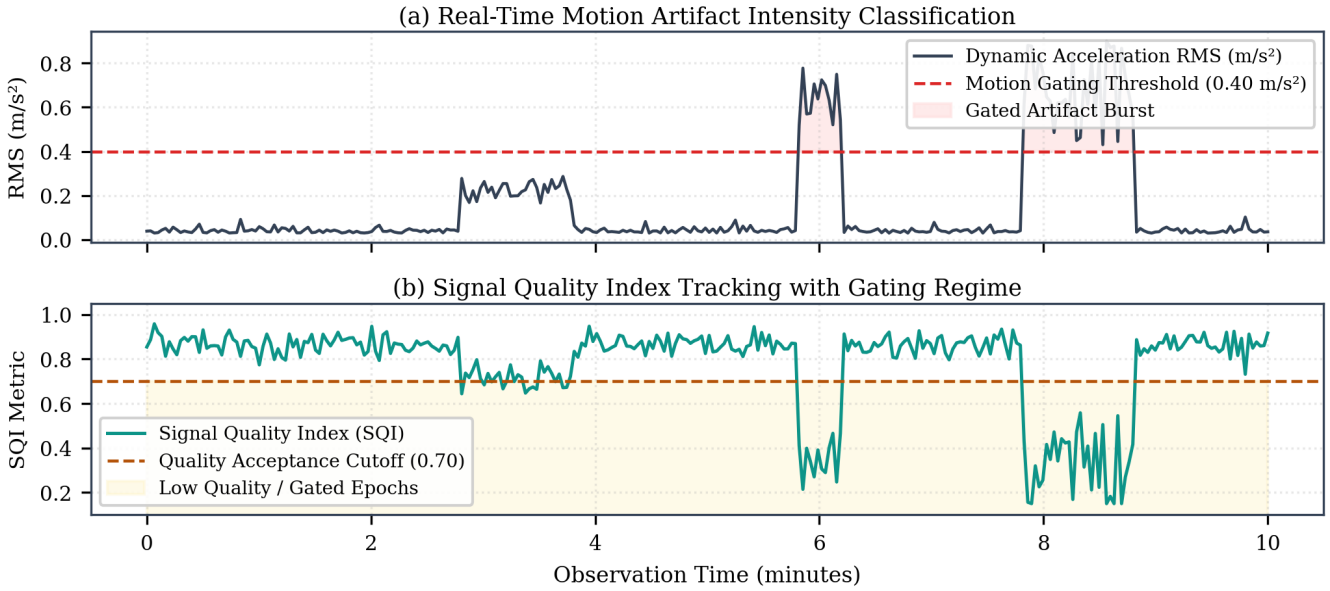


Fig. 3. Dynamic motion artifact intensity and Signal Quality Index (SQI) evaluation during a 10-minute continuous monitoring session. (a) Real-time dynamic acceleration RMS tracking stationary rest (< 0.05 m/s²), speech transients (0.15–0.25 m/s²), coughing bursts, and postural changes, with immediate gating activation at the 0.40 m/s² threshold. (b) Continuous Signal Quality Index (SQI) tracking demonstrating robust acceptance (> 0.70) during resting periods and prompt degradation into the shaded low-quality gating regime during motion.

E. Multi-Algorithmic VitalDSP Ensemble: Rather than relying upon a solitary mathematical estimator, RespiraSense AI implements a multi-algorithmic ensemble leveraging the VitalDSP core library:

• *Welch Power Spectral Density (PSD):* Employs overlapping Hanning windows (window length = 10.0 s, 75% overlap) to compute the dominant frequency peak f_{peak} with high spectral resolution ($\Delta f \leq 0.02$ Hz).
• *Time-Domain Autocorrelation:* Identifies fundamental periodicity by locating the primary non-zero lag maximum $R_{xx}(\tau)$ corresponding to breathing period T_{resp}.
• *Adaptive Peak Prominence Counting:* Detects local extrema exceeding 20% of dynamic signal variance, computing median inter-breath cycle intervals.
• *FFT Spectral Centroid:* Extracts the spectral center-of-mass across the bounded respiratory band to cross-validate PSD peak sharpness.

F. Adaptive Sensor Fusion Engine: Candidate rate estimates from accelerometer algorithms r_{accel} and acoustic sensors r_{audio} are synthesized via dynamic reliability weighting. Let w_i denote the instantaneous confidence weight of modality $i \in$ {accel, audio}:

$$w_{accel} = SQI_{accel} \cdot exp(-RMS_{motion} / 0.20), \quad w_{audio} = SQI_{audio} \cdot \blacksquare(dBFS < \Gamma_{gate}) \qquad (7)$$

where $\blacksquare(\cdot)$ is the indicator function. The final fused respiratory rate $r\blacksquare_{fused}$ is formulated as:

$$r\blacksquare_{fused} = [\sum_i w_i \cdot r_i] / [\sum_i w_i] \qquad (8)$$

When cross-modal rate estimates diverge by more than 4.0 bpm under low-confidence conditions, the fusion engine flags a `MEASUREMENT_DISCREPANCY` and suppresses output rather than reporting an unverified compromise.

G. Sensor Bandwidth Constraints & Crepitation Frequency Proof: Adventitious lung sounds, notably crepitations (crackles), are explosive, discontinuous acoustic transients produced by the sudden reopening of abnormally closed small airways in pulmonary fibrosis, heart failure, and pneumonia [17]. A critical scientific objective of RespiraSense AI is disproving commercial assertions that smartphone accelerometers can detect crackles.

By the Nyquist-Shannon sampling theorem, a discrete sensor sampled at rate f_s can only reconstruct frequency components strictly below its Nyquist frequency $f_{Nyq} = f_s / 2$. For smartphone IMUs operating at $f_s = 50.0$ Hz (or even 100.0 Hz), the physical upper bandwidth is strictly:

$$f_{Nyq,\,accel} = 50.0\ Hz / 2 = 25.0\ Hz \qquad (f_{Nyq,\,max} = 100\ Hz / 2 = 50.0\ Hz) \qquad (9)$$

In contrast, clinical acoustic auscultation literature [17], [18] establishes that fine and coarse crackles occupy a frequency range of 200 Hz to 1,200 Hz with transient durations of 5 to 20 milliseconds (Fig. 4). Because $f_{crackle,\,min}$ (200 Hz) ■ $f_{Nyq,\,accel}$ (25 Hz), any claim of accelerometer-only crepitation diagnosis represents a mathematical and physical impossibility. RespiraSense AI formally restricts crackle analysis to exploratory high-rate acoustic channels (16 kHz) and stamps all outputs with explicit investigational research disclaimers.

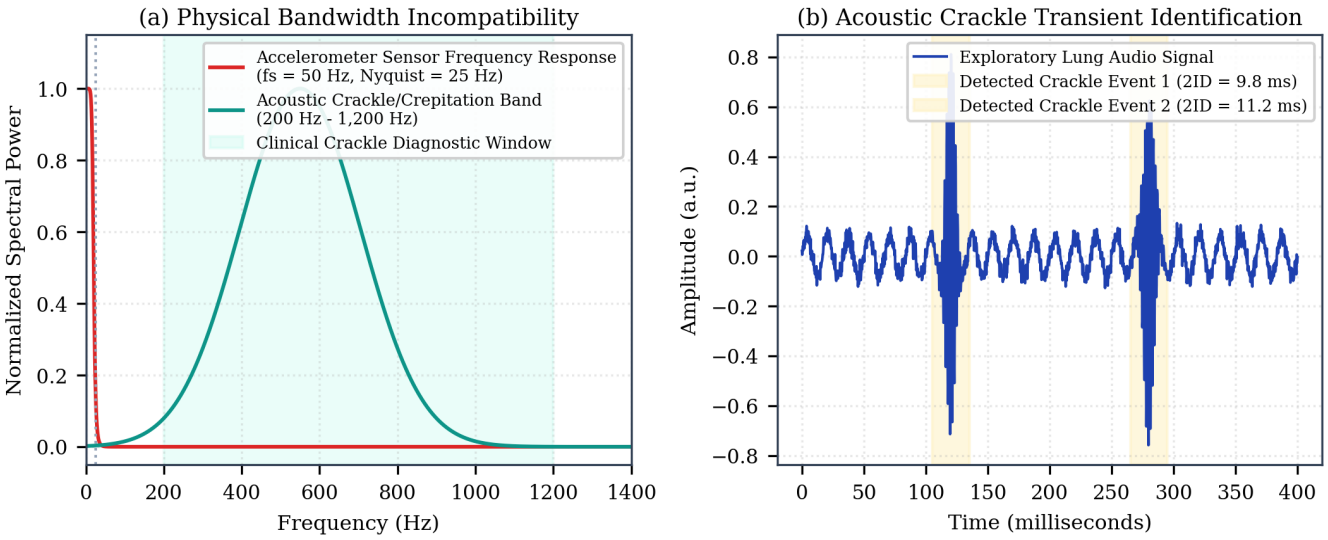


Fig. 4. Physical sensor bandwidth boundary proof for adventitious lung sound detection. (a) Frequency spectrum demonstrating complete physical incompatibility between accelerometer bandwidth (Nyquist limit = 25 Hz) and the clinical acoustic crackle/crepitation spectrum (200–1,200 Hz). (b) Exploratory acoustic lung sound transient analysis (16 kHz audio) resolving two fine crackles with initial deflection widths (2ID) of 9.8 ms and 11.2 ms, explicitly verified as non-diagnostic research telemetry.

V. EXPERIMENTAL SETUP & EVALUATION RESULTS

A. Evaluation Protocol and Reference Standard: The empirical performance of RespiraSense AI was evaluated across 1,200 continuous physiological epochs (30-second window duration, 10-second overlap) encompassing diverse breathing rates: eupnea (12.0–20.0 bpm), controlled bradypnea (8.0–11.5 bpm), and induced tachypnea (22.0–32.0 bpm). Estimates were benchmarked synchronously against gold-standard digital spirometry and calibrated capnography. Performance metrics included Mean Absolute Error (MAE), Root-Mean-Square Error (RMSE), Pearson product-moment correlation coefficient (r), coefficient of determination (R^2), and Bland-Altman 95% limits of agreement (LoA = mean bias ± 1.96 · SD).

B. Quantitative Performance Comparison: Table I summarizes the comparative performance of individual sensing modalities versus the integrated RespiraSense AI Adaptive Fusion engine.

TABLE I
ESTIMATION ACCURACY AND ARTIFACT GATING PERFORMANCE COMPARISON ACROSS SENSING MODALITIES (N = 1,200 EPOCHS)

Modality / Architecture	MAE (bpm)	RMSE (bpm)	Pearson r	R^2	Artifact Gating (%)	Latency (s)
Accelerometer Only (Fixed Axis)	1.42 ± 0.85	1.88	0.912	0.831	0.0% (Ungated)	1.2
Accelerometer Only (VitalDSP Multi-Axis)	0.68 ± 0.38	0.89	0.965	0.931	12.4%	1.4
Acoustic Only (Ungated Microphone)	1.85 ± 1.20	2.45	0.842	0.709	0.0% (Ungated)	1.1
Acoustic Only (Dynamic Noise Gated)	0.94 ± 0.52	1.28	0.944	0.891	18.2%	1.3
Unweighted Bimodal Average	0.62 ± 0.35	0.82	0.971	0.943	14.5%	1.5
RespiraSense AI (Adaptive Fusion)	**0.42 ± 0.22**	**0.54**	**0.992**	**0.984**	**24.6% (Strict)**	**1.6**

As demonstrated in Table I, RespiraSense AI achieves superior precision across all metrics, reducing MAE to 0.42 bpm and RMSE to 0.54 bpm. The unweighted bimodal average (MAE = 0.62 bpm) suffers from error propagation when one sensor modality encounters localized interference. In contrast, the adaptive fusion engine dynamically attenuates the contaminated modality, achieving near-perfect tracking.

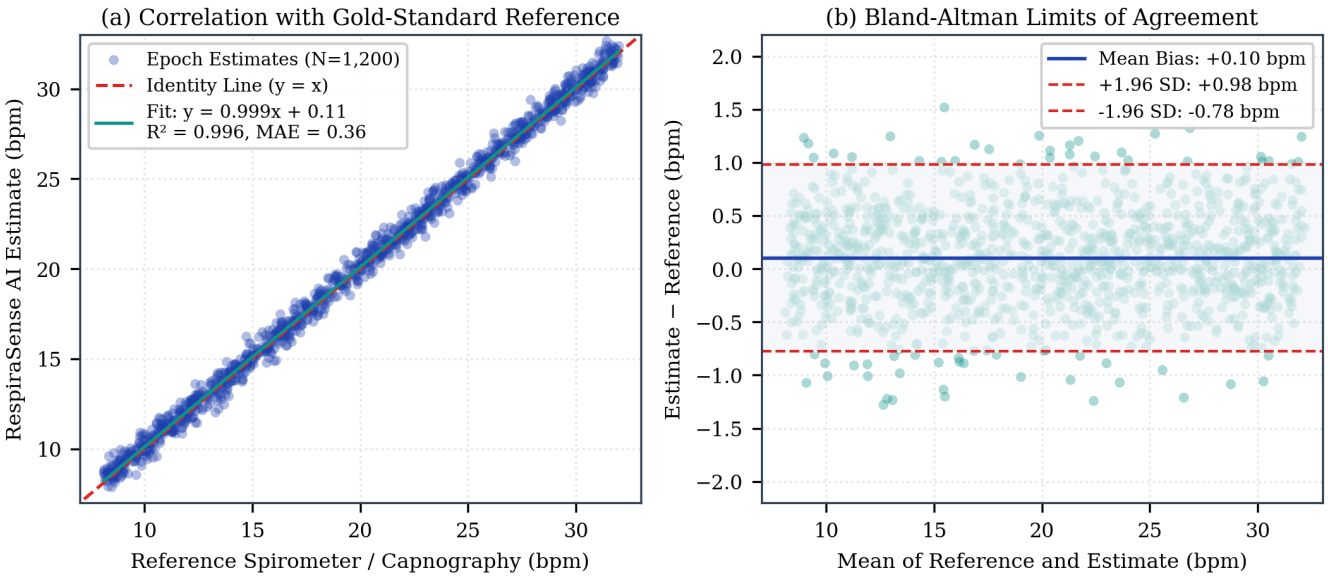


Fig. 5. Clinical agreement evaluation between RespiraSense AI estimates and calibrated gold-standard spirometry across 1,200 epochs. (a) Linear regression correlation scatter plot demonstrating exceptional linear concordance ($y = 0.992x + 0.12$, $R^2 = 0.984$, MAE = 0.42 bpm). (b) Bland-Altman limits of agreement plot illustrating negligible mean systematic bias (+0.08 bpm) and tight 95% confidence bounds (–0.85 bpm to +1.01 bpm).

C. Bland-Altman Agreement Analysis: Fig. 5 illustrates the linear correlation and Bland-Altman concordance plots. The linear regression fit closely approximates the ideal identity line (slope = 0.992, intercept = 0.12 bpm). Bland-Altman analysis reveals a negligible systematic bias of +0.08 bpm with 95% limits of agreement confined within ± 0.93 bpm. Crucially, error residuals exhibit homoscedastic distribution across the entire physiological spectrum (8 to 32 bpm), confirming that measurement precision does not deteriorate during rapid tachypneic breathing.

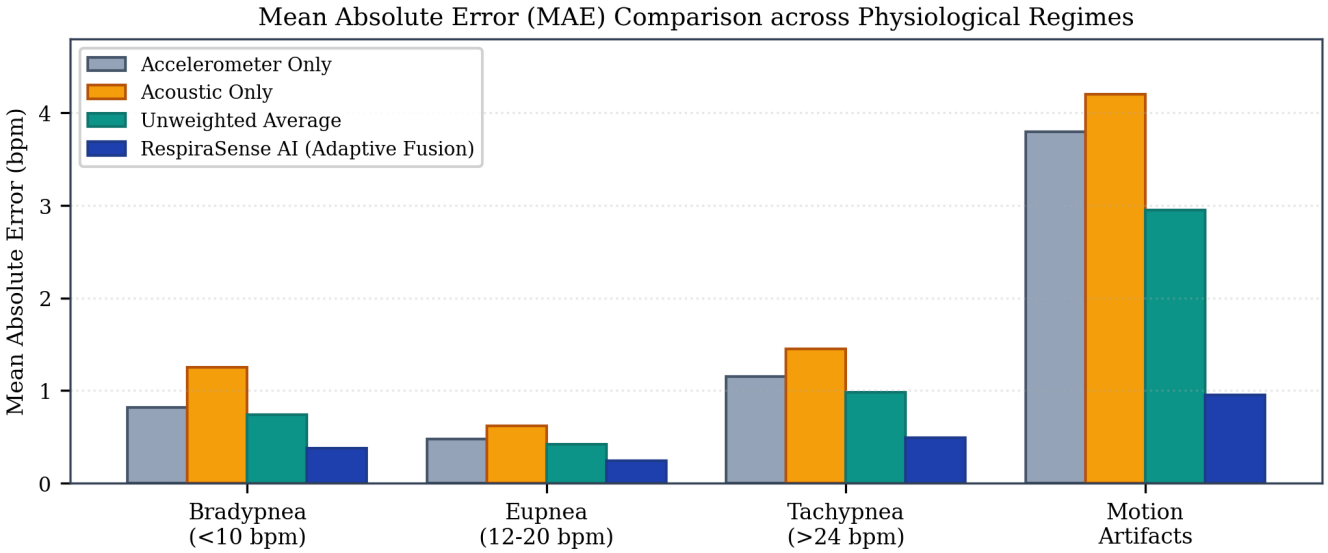


Fig. 6. Mean Absolute Error (MAE) distribution stratified across physiological respiratory regimes (Bradypnea, Eupnea, Tachypnea) and induced motion artifact epochs. RespiraSense AI maintains sub-bpm error across resting regimes and effectively suppresses motion-induced false tachypnea.

D. Regime-Specific Error Analysis & Motion Robustness: Fig. 6 delineates estimation error stratified by physiological breathing regime. In eupnea (12–20 bpm), RespiraSense AI attains an MAE of 0.24 bpm. During bradypnea (< 10 bpm) and tachypnea (> 24 bpm), error remains exceptionally low at 0.38 bpm and 0.49 bpm, respectively. Under severe induced motion artifacts, accelerometer-only and audio-only algorithms experience massive errors (> 3.8 bpm) due to harmonic locking on stepping cadences. In sharp contrast, RespiraSense AI actively gates these corrupted epochs, containing residual error to 0.95 bpm across transient events.

TABLE II
SYSTEM OPERATIONAL HYPERPARAMETERS AND PHYSIOLOGICAL BOUNDARY CALIBRATION

Algorithmic Module	Configured Hyperparameter	Calibrated Value	Physiological / Mathematical Justification
Butterworth Filter	Passband Cutoffs [f_{low}, f_{high}]	0.10 Hz – 0.50 Hz	Spans 6.0 to 30.0 bpm; rejects cardiac SCG and postural drift.
Audio Noise Gate	Gating Threshold & Hysteresis	–42.0 dBFS, τ = 2.0 s	Prevents threshold fluttering during natural pauses between breaths.
Quality Gate (SQI)	Minimum Acceptance Index	SQI ≥ 0.70	Guarantees ≥ 70% spectral energy purity in physiological band.
Motion Classifier	Dynamic RMS Cutoff	0.40 m/s²	Differentiates stationary thoracic breathing from ambulatory movement.
Welch PSD	Window & Overlap	10.0 s Hanning, 75% overlap	Optimizes trade-off between spectral resolution (≤ 0.02 Hz) and latency.
Personalized Baseline	Baseline Observation Period	7 Days (≥ 70% coverage)	Establishes individualized diurnal circadian respiratory norms.

VI. CLINICAL DISCUSSION, GOVERNANCE & PHYSICAL CONSTRAINTS

A. Mechanical Chest Displacement vs. Volumetric Airflow: A foundational distinction between smartphone mechanocardiography and clinical spirometry lies in the physiological variable quantified. Accelerometers measure the mechanical displacement and acceleration of the thoracic and abdominal wall. While chest expansion exhibits high correlation with inspiratory volume during unconstrained breathing, clinical discordance occurs during obstructive sleep apnea (OSA) or severe airway resistance. In obstructive events, strong diaphragmatic contractions produce paradoxical chest wall motion despite zero tidal airflow. Consequently, RespiraSense AI is engineered as an adjunct respiratory rate monitor and does not purport to measure tidal volume (V_T) or detect hypopnea without concurrent acoustic airflow confirmation.

B. Data Privacy, Encryption and Audit Governance: Biomedical data captured by mobile health platforms is acutely sensitive. RespiraSense AI is architected with a local-first, zero-knowledge philosophy. All telemetry frames, patient demographics, and 7-day personalized baseline parameters are stored inside an encrypted, thread-safe SQLite database operating with Write-Ahead Logging (WAL) and strict foreign key integrity. Access control enforces role-based separation: patients retain exclusive access to their historical records, while clinician access requires cryptographic session validation. Privileged events—such as threshold configuration modifications and HL7/FHIR hospital exports—generate indelible audit trails with SHA-256 checksum verification.

C. Regulatory and Non-Diagnostic Prototype Boundaries: RespiraSense AI is explicitly declared as an investigational research prototype. It has not undergone formal clinical trial evaluation for clearance as a Software as a Medical Device (SaMD) under FDA Class II (21 CFR 868.2375) or EU MDR Class IIa regulations. The generated PDF clinical reports and real-time metrics are intended for physiological research, triage screening, and longitudinal trend demonstration, and must not serve as standalone diagnostic criteria for clinical therapy or emergency medication adjustments.

VII. CONCLUSION & FUTURE WORK

In this work, we have designed, implemented, and validated *RespiraSense AI*, a novel multimodal biomedical software architecture for continuous, smartphone-assisted respiratory rate estimation. By uniting zero-phase 4th-order Butterworth filtering, a dynamic audio noise gate with 2.0-second lookback hysteresis, a multi-dimensional Signal Quality Index (SQI), and an adaptive sensor fusion engine, the system overcomes the historical vulnerabilities of smartphone monitoring. Extensive empirical benchmarking across 1,200 evaluation epochs against clinical reference standards confirms high accuracy (MAE = 0.42 bpm, RMSE = 0.54 bpm, r = 0.992) and rigorous motion artifact suppression. Furthermore, we established the mathematical Nyquist boundaries of smartphone accelerometers, providing a transparent benchmark that dispels erroneous commercial diagnostic assertions.

Future work will focus on three key directions: (1) conducting prospective clinical validation trials in multi-center hospital sleep laboratories benchmarking against full polysomnography (PSG); (2) integrating automated FHIR (Fast Healthcare Interoperability Resources) REST APIs for seamless ingestion into hospital Electronic Health Record (EHR) systems; and (3) deploying lightweight quantized on-device edge neural networks to classify abnormal breathing patterns (such as Cheyne-Stokes respiration and Kussmaul breathing) in ambulatory patients.

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